

Atlas Coaching Intelligence Architecture

Core design principles

Atlas should be built as a **hybrid coaching system**, not as a single language model with a fitness persona. The evidence base on resistance training is strong enough that many high-impact decisions can be standardized: resistance training clearly improves strength, hypertrophy, power, endurance, gait speed, balance, and function; higher loads tend to maximize strength gains; hypertrophy can be achieved across a broad loading spectrum when effort is sufficient; and program outcomes depend heavily on variables such as volume, frequency, rest intervals, and progression over time. That makes progression logic, fatigue management, safety thresholds, trend detection, and substitution rules good candidates for deterministic systems rather than free-form text generation. ¹

Atlas should also assume that **consistency and adherence matter more than program cleverness for most users**. ACSM's 2026 update emphasized that the biggest gains come from moving from no resistance training to regular resistance training, and that individualization, enjoyment, and sustainability matter more than chasing complex "perfect" programming. In parallel, behavior-change literature consistently links autonomy support, goal setting, monitoring, feedback, and self-determined motivation with better exercise adherence. In practice, that means Atlas's intelligence has to model not just training stress, but also schedule friction, motivation, confidence, and user preference. ²

The system should reason around **adaptation, fatigue, and risk as interacting time-series problems**. Sleep loss, travel, illness, concurrent endurance work, and intensified training can all degrade performance or recovery, but no single signal is specific enough to explain why. Reviews on athlete monitoring consistently show that subjective wellness, session-RPE, performance trends, and physiological markers such as HRV are useful, but each is noisy when interpreted in isolation. Atlas therefore needs a multi-signal readiness model, a causal hypothesis layer, and a confidence system that decides when to ask for clarification instead of pretending certainty. ³

Finally, Atlas should behave like a good coach in conversation: it should explain *why*, reference user history, calibrate confidence, and avoid empty praise. This is not just a UX preference. Autonomy-supportive, competence-building communication is strongly aligned with the motivational mechanisms that predict exercise persistence. A world-class AI coach should therefore separate **decision generation** from **decision explanation**: the rules engine decides, the LLM translates those decisions into clear, specific, history-aware coaching language. ⁴

Complete coaching knowledge taxonomy

Athlete model and constraints. This domain matters because every good program depends on who the person is, what they want, what they can do safely, and what environment they train in. Atlas should know age, sex if provided, training age, injury history, medical red flags, equipment access, schedule constraints, recovery capacity, sleep patterns, preferred exercises, disliked exercises, and goal hierarchy. Deterministic

logic should handle hard constraints such as equipment mismatch, contraindicated movements, session duration caps, and medical red flags. LLM reasoning should handle ambiguous preference tradeoffs, coaching questions, and clarification when the user's stated goal conflicts with their behavior. Required user data include demographics, goals, environment, and constraints; historical data include attendance, skipped sessions, substitutions, adherence by training day, and pain flags. Permanent storage should include stable constraints and preferences; transient lifestyle disruptions should be inferred or updated at runtime. This architecture is justified because adherence, long-term development, and safety all depend on individual context rather than generic prescriptions. 5

Training adaptation science. This is the core physiology layer of the coach. Atlas should know progressive overload, hypertrophy mechanisms, neural and morphological strength adaptations, muscular endurance, power, rate of force development, velocity-specific adaptation, specificity, periodization, deloading, tapering, and detraining. Deterministic logic should govern progression bands, weekly volume accounting, intensity distribution, exposure frequency, taper/deload triggers, and goal-specific templates. LLM reasoning should be reserved for edge-case synthesis, user education, and explaining tradeoffs between competing goals. Required user data include exercise selection, loads, reps, RIR/RPE, frequency, and goal phase; historical data include per-muscle weekly set volume, estimated strength trajectories, proximity-to-failure history, and phase response by goal. Permanent storage should include the adaptation ontology and user response tendencies; current phase readiness should be inferred at runtime. This domain is supported by ACSM position stands and meta-analyses showing dose-response relationships for loading, set volume, and frequency, plus evidence that periodization is especially useful for trained lifters and strength-oriented goals. 6

Exercise ontology and biomechanics. Atlas needs a structured representation of exercises because "bench press" is not just a name; it is a relationship between movement pattern, muscles trained, stabilization demand, range of motion, resistance profile, fatigue cost, and technique requirements. Atlas should store for each exercise: movement pattern, primary/secondary movers, stabilizers, unilateral/bilateral status, axial loading, spinal demand, shoulder demand, local fatigue cost, systemic fatigue cost, technical complexity, range-of-motion demands, progression options, regression options, substitution graph, and injury cautions. Deterministic logic should power substitution, equipment-aware alternatives, movement-balance checks, fatigue indexing, and PR normalization within pattern families. LLM reasoning should explain exercise choice and guide form conceptually, but should not invent movement metadata. Required user data include limb limitations, pain regions, available equipment, mobility constraints, and exercise history. Historical data should include exercise familiarity, technique quality flags, progression slope, pain history by exercise, and substitution success. Permanent storage should include exercise metadata and user-specific tolerances; daily movement quality and pain response should be inferred at runtime. The need for this domain is supported by biomechanics literature showing that squat, deadlift, bench press, and pulling variations impose distinct joint and muscle demands, and by ACSM's finding that many tools and modalities can work when the underlying movement and effort demands are matched. 7

Programming and session design. Elite coaches do not only pick exercises; they organize them into productive sessions and mesocycles. Atlas should know volume, intensity, frequency, exercise order, warm-ups, rest intervals, supersets, giant sets, drop sets, cluster sets, pauses, tempo, AMRAPs, repetition limits, and when to use or avoid failure. Deterministic logic should handle weekly set accounting, session ordering, warm-up ramping, rest-time defaults, advanced-technique eligibility, and session density control. LLM reasoning should handle session summaries, why a method was used, and how to adjust when the user's context changes unexpectedly. Required user data include available time, goal priority, equipment, and

recovery status; historical data include response to rest lengths, tolerance for density techniques, session completion rates, and whether advanced methods improved or harmed adherence. Permanent storage should include exercise-method compatibility and user preferences; event-specific session density decisions should be inferred at runtime. This domain matters because rest length, set configuration, exercise order, and load/intensity meaningfully affect strength, hypertrophy, power, and training sustainability. ⁸

Progression and autoregulation. This is where Atlas becomes a coach instead of a tracker. Atlas should know how to increase load, reps, or sets; when to hold or regress; how to use RPE, RIR, velocity, and estimated capability; how to identify noise versus signal; and how to decide whether a bad session should be ignored, interpreted, or escalated. Deterministic logic should own progression decisions, expected-performance models, plateau detection, and trend alarms. LLM reasoning should generate the explanation and ask for missing context when confidence is low. Required user data include performed load, reps, set count, RIR/RPE, optional velocity, pain, sleep, and session comments; historical data include progression slope by exercise, performance variability, readiness relationships, and previous plateau resolution strategies. Permanent storage should include user-specific progression response curves and exercise baselines; readiness-adjusted expected performance should be inferred at runtime. This domain is strongly supported by literature validating RPE/RIR and movement velocity as practical tools, and by evidence that frequent training-to-failure lengthens recovery and may not offer meaningful hypertrophy advantages over non-failure training when volume is matched. ⁹

Fatigue, recovery, and readiness. Atlas must understand central fatigue, peripheral fatigue, local muscle fatigue, systemic fatigue, sleep, nutrition, stress, age, training age, concurrent training, travel, illness, and heat because the same missed reps can mean very different things in different contexts. Deterministic logic should aggregate readiness from multiple signals, estimate local and systemic fatigue, and trigger recovery-preserving changes. LLM reasoning should formulate causal hypotheses and ask the best next question when the score is low-confidence. Required user data include sleep, soreness, motivation, stress, illness symptoms, travel, heat exposure, endurance training, hydration status if available, and pain. Historical data should include recovery half-life by muscle and movement pattern, response to failure training, prior sleep-performance relationships, travel sensitivity, and post-illness return patterns. Permanent storage should include baseline variability and known sensitivities; current fatigue state should be inferred at runtime. This domain matters because fatigue and recovery kinetics differ by exercise type, body region, age, effort level, and outside-life stressors, and because single recovery metrics are not specific enough to stand alone. ¹⁰

Safety, injury screening, and minimal nutrition support. Atlas is primarily a training coach, but it still needs enough safety and nutrition knowledge to avoid bad advice. Atlas should know basic protein and energy balance, hydration, supplement basics with strong evidence, and recovery nutrition. It should also know injury-warning patterns, exertional heat risk, concussion red flags, fever/viral illness rules, and when symptoms warrant medical evaluation rather than coaching. Deterministic logic should own stop/modulate/escalate decisions, pain thresholds, symptom-triggered training reductions, and supplement guidance limited to low-risk basics. LLM reasoning should explain the recommendation, communicate boundaries, and avoid pretending to diagnose. Required user data include body-mass trend, goal phase, dietary pattern, illness symptoms, pain location, symptom onset, and red-flag responses. Historical data include recurrent pain patterns, prior successful modifications, and body-weight changes relative to goal. Permanent storage should include stable injury history and supplement preferences; current illness/injury severity should be inferred at runtime. This domain is supported by protein and nutrient-timing position

stands, hydration guidance, and medical guidance on fever, exertional symptoms, concussion, and serious musculoskeletal red flags. ¹¹

Behavior change, coaching psychology, and communication. This domain determines whether users keep showing up long enough for the physiology to matter. Atlas should know habit formation, self-efficacy, goal setting, autonomy support, accountability, identity-based change, motivational interviewing style, and how experienced coaches use specific—not generic—feedback. Deterministic logic should control reinforcement cadence, streak logic, reminder timing, praise quality filters, and anti-repetition constraints. LLM reasoning should personalize encouragement, summarize trends, ask smart follow-up questions, and explain the *why* behind decisions in a tone calibrated to the user. Required user data include preferred coaching style, response to reminders, motivational triggers, confidence, and friction points. Historical data include drop-off patterns, effective message styles, missed-workout reasons, and which explanations changed downstream behavior. Permanent storage should include a coaching-style profile; moment-to-moment emotional state should be inferred at runtime. This domain matters because autonomy, competence, and meaningful feedback are durable predictors of exercise adherence, whereas repetitive generic motivation tends to decay in value quickly. ¹²

Training, exercise, and programming ontology

Atlas should treat exercise knowledge as a **two-axis graph**. The first axis is **movement pattern**. The second axis is **resistance implementation**. Pattern tells Atlas what functional and muscular role an exercise plays; implementation tells Atlas how the movement is loaded, stabilized, progressed, and substituted. This is a better schema than grouping by exercise name alone because many reasonable substitutions preserve the pattern while changing implement, stability demand, or joint stress. ACSM's recent guidance that barbells, bands, bodyweight, machines, and other nontraditional tools can all produce meaningful benefits supports this pattern-first approach. ¹³

Squat pattern. Atlas should store knee-dominant versus hip-dominant squat variants, depth demands, torso angle, unilateral contribution, and axial loading. The primary movers are typically quadriceps and gluteals, with trunk musculature contributing heavily to stabilization; depth, stance, trunk position, and tibia angle meaningfully change joint loading and muscular demands. Squats usually generate moderate-to-high local fatigue and moderate systemic fatigue, with higher technical complexity for free-weight bilateral barbell variants than for machine-based versions. Progressions include load, reps, depth, tempo, pauses, and variant difficulty; regressions include box squats, goblet squats, supported split squats, and machine or bodyweight variants. Common faults include collapsing positions, uncontrolled descent, and depth/stance choices mismatched to user mobility or tolerance. ¹⁴

Hinge pattern. Atlas should distinguish conventional deadlift-style hinges, Romanian deadlifts, hip thrusts, back extensions, kettlebell swings, and machine hinges because hip-extension demand, spinal loading, and ballistic intent differ meaningfully. Hip extensors and posterior-chain tissues are primary, with erector spinae and other trunk musculature often carrying large stabilization demands; conventional deadlifts tend to generate high systemic fatigue, while supported machine or isolated hip-extension options usually generate less. Progression can occur through load, leverage, range, pauses, accommodating resistance, or ballistic intent depending on goal. Regressions include trap-bar pulling, rack pulls, dumbbell RDLs, cable pull-throughs, and hip bridges. ¹⁵

Press pattern. Atlas should split horizontal and vertical pressing, plus free-weight, machine, cable, and bodyweight forms. Primary movers usually include pectoralis major, anterior deltoid, and triceps in horizontal pressing, with greater shoulder-elevation demands in overhead work. Technical complexity and shoulder tolerance vary widely with grip width, bench angle, shoulder position, and implement freedom. Progressions include load, reps, ROM, pause work, incline/bench angle, and unilateral versions; regressions include machines, neutral-grip dumbbells, push-up progressions, and partial-ROM options when tolerated.

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Pull pattern. Atlas should represent vertical pulls, horizontal rows, scapular control demands, grip options, and trunk support. Primary movers typically include latissimus dorsi, upper back musculature, and elbow flexors, with supported rows generally producing lower systemic cost than unsupported or bodyweight vertical pulling. Progressions include added load, ROM, pauses, tempo, grip changes, and unilateral variants; regressions include pulldowns, chest-supported rows, cable work, or bands.

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Carry, rotation, anti-rotation, and locomotion. These categories matter because they broaden Atlas beyond bodybuilding splits and let it coach general fitness, athletic performance, and trunk function. Carries and locomotion patterns usually produce moderate systemic fatigue with relatively low technical barriers, and they are useful for conditioning, grip, posture, work capacity, or low-skill loaded movement. Rotation and anti-rotation work often have lower absolute loading but high value for trunk control, sport transfer, and lower-fatigue accessory programming. Because the direct literature base here is thinner and more context-dependent than for squats, presses, and deadlifts, Atlas should keep their progression and substitution logic more conservative and pattern-based.

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Unilateral and isolation work. Atlas should treat unilateral work as a modifier layered onto squat, hinge, press, or pull patterns, not as a separate kingdom. Its value is often coordination, side-to-side loading management, joint tolerance, and fatigue distribution. Isolation work should be modeled as high-local-fatigue, low-systemic-fatigue tools that are valuable for hypertrophy volume, rehab-compatible training, or when systemic recovery is constrained. ACSM's 2026 overview and the broader hypertrophy literature support the idea that many programming configurations can work, so the key is not whether an exercise is "functional" or "isolated," but what role it plays in the total program and fatigue budget.

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Implementation layer. Barbells usually allow the highest absolute loading but often demand more skill and stabilization. Dumbbells allow freer joint paths and simpler bilateral asymmetry management. Machines reduce stabilization demands and can be very useful for hypertrophy volume or injury-sensitive users. Cables usually expand resistance-direction options and fine-grained load control. Kettlebells are especially useful for ballistic and offset-loading patterns. Bodyweight is highly accessible but can have awkward progression ceilings in some movements. Bands are viable tools, but their resistance profile must be understood when comparing them to gravity-based options. Atlas should therefore never store "exercise = muscles"; it should store "exercise = movement pattern + implementation + resistance profile + fatigue cost + technical demand + substitution neighborhood." ACSM's 2026 guidance that equipment type is often less important than consistent effort strongly supports this abstraction.

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Programming should be represented in a similarly structured ontology. **Volume** should be stored at both exercise level and estimated per-muscle level, because hypertrophy decisions often care about muscle-specific dose while strength decisions often care about lift-specific practice. **Intensity** should be stored as both external load and internal effort, because the load that maximizes strength is not identical to the load that can maximize hypertrophy. **Frequency** should be tracked per movement and per muscle, because

increasing weekly frequency can be useful when it improves quality, adherence, or volume distribution. **Exercise order** should prioritize what matters most in the session, especially when strength in multi-joint lifts is a goal. **Warm-ups** should be treated as performance preparation, not random cardio. **Rest intervals** should be goal-dependent, with longer rests usually favored for maximal strength and often helpful for preserving hypertrophy-relevant volume quality. **Supersets** and **drop sets** are best treated as time-efficiency or density tools, not universal upgrades; **cluster sets** are especially useful when Atlas wants to preserve velocity and quality under moderate-to-high load. **Tempo** and **pauses** should be prescription modifiers, not default complexity. **AMRAPs** should be used sparingly as diagnostics or controlled overreach, not as baseline programming. **RPE/RIR** and optional **velocity** are the main autoregulation inputs. ²¹

Progression, recovery, personalization, and safety logic

The progression engine should operate like a **state machine**, not like a vibe. Its job is to classify each exercise exposure as productive, ambiguous, fatigued, or risky; update an expected-performance model; and then choose the lowest-complexity intervention that plausibly keeps adaptation moving. This is the part of Atlas that should be highly deterministic because users will notice inconsistency immediately if the model improvises load jumps or deloads from one day to the next. Deterministic control is especially justified by the evidence that load, volume, rest, failure, and frequency have patterned effects, while readiness metrics are helpful but noisy and require trend-based interpretation. ²²

A practical deterministic progression framework for most strength and hypertrophy work would look like this:

Session outcome	Deterministic interpretation	Default action
Hit all planned sets, reached top of rep target, technique acceptable, pain low, and actual effort easier than prescribed	Underloaded relative to current capability	Increase load next exposure if smallest increment still keeps user within target rep/RIR band
Hit all planned sets with appropriate effort and technique	On target	Maintain load or progress by a small rep increase
Missed top of range but stayed within minimum reps and within tolerance bands	Normal variability	Hold progression; do not react to one mediocre exposure
Missed minimum reps on multiple sets, effort much harder than prescribed, or velocity meaningfully suppressed	Likely fatigue or overprescription	Hold or reduce load, or reduce sets if recovery signals are also poor
Local joint pain, compensatory technique, or sharp focal pain emerged	Elevated injury risk	Switch exercise or reduce ROM/intensity immediately
Multiple consecutive exposures underperform expected output despite acceptable recovery inputs	Candidate plateau or bad programming	Change exercise variant, rep range, volume distribution, or phase goal

Atlas should generally **increase weight before sets**, and **increase reps before sets** when the goal is simple progression within a rep range. Sets are the expensive lever because they escalate fatigue and time cost. Atlas should therefore reserve set increases for cases where performance has stabilized, adherence is good, recovery is acceptable, and the modeled stimulus appears too small for the goal. This direction is consistent with the broad evidence that volume has dose-response benefits but also diminishing returns, and that more is not automatically better if it degrades performance quality or recovery. ²³

Atlas should **ignore one bad session** unless it also contains red flags. Sleep loss, travel, acute stress, illness, or even poor warm-up quality can depress performance transiently. Reviews on sleep deprivation, travel fatigue, athlete monitoring, and illness all support trend-based interpretation over single-point reactions. A true plateau is better defined as **repeated underperformance against a realistic expected-performance curve across multiple exposures of the same movement family**, with acceptable technique and adherence, and without obvious explanation from sleep, illness, or overload. ²⁴

To distinguish poor sleep, fatigue, injury, under-recovery, bad programming, illness, or low motivation, Atlas should use a **causal triage matrix**:

Pattern	Atlas should suspect	Strongest supporting signals
One- to two-day global performance dip, poor sleep report, low readiness, otherwise normal pain	Poor sleep or transient systemic fatigue	Sleep duration/quality down, motivation may be normal, no focal pain
Multi-day soreness, flat motivation, reduced performance after several hard sessions or failure work	Under-recovery / accumulated fatigue	High recent load, high proximity-to-failure exposure, soreness, suppressed readiness
Sharp focal pain, ROM loss, swelling, limping, or compensation	Injury risk rather than normal fatigue	Pain localized to tissue/joint, worsens with specific movement, function drops
Low energy plus fever, chest symptoms, body aches, unusual shortness of breath	Illness	Fever/viral symptoms, elevated resting strain, global malaise
Multiple stalls in one movement family despite good recovery and attendance	Bad programming or stale stimulus	Plateau isolated to pattern, no obvious systemic cause
Normal physiological readiness but repeated skipped sessions, lower enthusiasm, short or rushed workouts	Motivation or friction problem	Time pressure, schedule conflict, boredom, low engagement comments

That logic is evidence-aligned. Sleep restriction can reduce force output, especially across repeated nights. Training to failure prolongs recovery relative to non-failure work. Travel fatigue and jet lag can impair readiness and sleep. Illness with fever is a clear reason to avoid normal training. Subjective wellness measures help detect fatigue, but they do not identify cause alone. ²⁵

Atlas should model fatigue on at least three levels. **Local muscle fatigue** is pattern- and muscle-specific and should drive per-muscle volume spacing. **Peripheral fatigue** is the exercise-induced reduction in force or power from mechanisms at or distal to the muscle and is especially relevant for high-volume,

metabolically costly work. **Central fatigue** is reduced voluntary activation and broader neural drive. In practice, Atlas should not pretend it can clinically separate them perfectly, but it should still represent them because their signatures differ. Lower-body compounds, repeated eccentric work, failure training, and older age tend to extend recovery windows; lower-body work often needs longer recovery than upper-body work, and older adults often need more conservative recovery assumptions even though they still benefit substantially from resistance training. ²⁶

For **deloads**, Atlas should trigger them deterministically from converging evidence, not as calendar superstition. Good triggers include repeated underperformance, high fatigue score, rising soreness, falling motivation, and recent high-load or high-volume accumulation. Deloads should usually reduce total demand by some combination of set volume, intensity, or frequency while preserving movement exposure. The current literature supports deloading as a purposeful reduction in demand, but also shows the direct evidence base is still growing and somewhat context-specific, which means Atlas should use conservative, explainable deload logic rather than pseudo-scientific certainty. ²⁷

For **tapering and peaking**, Atlas should narrow the target population. Beginners and general fitness users usually need recovery and consistency, not peaking. Powerlifters and athletes in test weeks or competition windows do need this logic. The evidence on strength tapering supports reducing volume substantially while maintaining relative intensity and lift specificity close enough to the event to preserve skill. Atlas should therefore keep a separate peaking engine for strength-sport users rather than polluting the general-population logic with unnecessary complexity. ²⁸

Personalization should change by **training level, goal, and life context**. Beginners benefit from simple progression, lower exercise complexity, high skill repetition, and broad movement coverage. Intermediates need more careful volume and fatigue management. Advanced lifters need tighter exposure control, more phase-specific variation, and stronger plateau logic because progress is slower and noise matters more. Powerlifters need lift specificity, competition-standard practice, peaking, and clearer fatigue stratification around the squat/bench/deadlift. Bodybuilders need better muscle-specific volume accounting, isolation libraries, and less emphasis on absolute load. Older adults need strength plus power and function, but often with more careful recovery assumptions and exercise selection. Youth athletes need supervised, age-appropriate progression with strong emphasis on skill and safety. Busy parents and constrained users need session-density tools, minimal-effective-dose templates, and highly flexible substitutions. Home-gym users need an equipment-aware graph. Commercial-gym users need congestion alternatives. ²⁹

Safety should use a **traffic-light system**.

State	Atlas action
Green	Coach normally; minor soreness, ordinary fatigue, no meaningful pain escalation
Yellow	Reduce intensity, volume, or ROM; modify exercise; ask targeted questions
Red	Stop the exercise or workout; advise medical evaluation when indicated

Green means ordinary training fatigue. Yellow means symptoms such as unusual soreness, mild but increasing pain, recent poor sleep, mild illness without fever, travel fatigue, or accumulating under-recovery. Red means chest pain, unusual shortness of breath, dizziness/fainting, palpitations with symptoms, fever/viral syndrome for normal training, concussion symptoms, inability to bear weight, obvious deformity,

severe swelling, numbness/tingling, loss of function, or rapidly escalating pain. Those are not coaching problems; they are triage problems. ³⁰

Data, memory, and knowledge base architecture

Atlas needs a memory system that distinguishes **facts**, **time series**, **derived traits**, **session context**, and **conversation context**. Many AI coaching systems fail because they dump everything into one amorphous memory store. Atlas should not do that.

A workable storage model is this:

Store	What belongs here	Persistency
Permanent user profile	Date of birth or age bracket, training age, goal hierarchy, sport, equipment, preferences, stable injuries/limitations, coaching style preference, optional body metrics	Long-term
Workout history	Exercise, variant, load, reps, sets, RIR/RPE, rest, tempo, notes, pain flags, completion status	Long-term
Long-term trends	Rolling volume by muscle/pattern, estimated strength capability, adherence rates, readiness baseline, body-weight trend, PR history, plateau history	Long-term, recomputed
Current session state	Planned workout, performed warm-ups, current performance, pain, readiness, time remaining, substitutions used	Ephemeral until session closes
Recent fatigue state	Last 7–21 days of sleep, soreness, stress, concurrent cardio, illness symptoms, travel, heat exposure, recent failure exposure	Short- to mid-term
Exercise memory	Familiarity, best tolerated variants, historical response, usual progression rate, typical technique faults, pain associations	Long-term
Coach memory	Effective communication style, reinforcement frequency, prompt-response patterns, what explanations help this user act	Long-term
Conversation memory	Recent questions, clarifications, unresolved ambiguities, temporary goals, session-specific context	Short-term

What should be stored permanently versus inferred at runtime is crucial. **Store** the things that are stable, identity-level, or repeatedly useful: goals, training age, preferred schedule, disliked exercises, equipment, injury history, successful substitutions, historic PRs, progression rate by movement, and user-specific sensitivities such as “sleep-sensitive bench performance” if repeatedly observed. **Infer at runtime** readiness, current fatigue, whether today’s underperformance is meaningful, whether a deload is needed, and whether the user likely needs encouragement, education, or direct instruction. Those are states, not traits. ³¹

The knowledge base itself should be structured like an internal coaching operating system:

```
/coaching_kb
  /adaptation_science
    hypertrophy
    strength
    power_rfd
    muscular_endurance
    periodization
    recovery
  /exercise_ontology
    movement_patterns
    exercise_entries
    equipment_profiles
    substitution_graph
    contraindication_rules
  /programming
    templates
    set_rep_schemes
    warmup_rules
    rest_interval_rules
    advanced_techniques
    taper_deload_logic
  /monitoring
    readiness_model
    fatigue_model
    expected_performance_model
    plateau_detection
    risk_detection
  /behavior_and_communication
    coaching_styles
    question_strategies
    explanation_patterns
    adherence_interventions
  /nutrition_minimums
    protein
    energy_balance
    hydration
    recovery_nutrition
    supplements_basic
  /safety
    stop_rules
    reduce_rules
    escalation_rules
    symptom_checkers
```

Atlas should also maintain a **user-specific derived layer** rather than rewriting the universal knowledge base every time someone trains. For example, the universal KB might say that longer rest intervals usually

support higher-strength performance and can help preserve quality volume for hypertrophy; the user-specific layer should say that *this* user’s bench performance degrades sharply when rest drops below ninety seconds, while *that* user tolerates antagonist supersets well. ³²

A strong memory architecture also requires **entity resolution**. Atlas should know that “low-bar squat,” “competition squat,” and “back squat” are related but not identical; that “DB incline press” is a press variant, not a completely unrelated exercise; and that “left shoulder annoyance on incline DB press” should update the shoulder-tolerance profile, not disappear into chat history. If Atlas cannot resolve these entities reliably, it will never become genuinely personalized.

Deterministic engine, LLM responsibilities, confidence system, evidence hierarchy, and roadmap

The deterministic engine should own the parts of coaching where **consistency, auditability, and safety** matter most. At minimum, that includes progression decisions, weekly set and exposure calculations, exercise substitutions, readiness and fatigue scoring, expected-performance modeling, plateau detection, deload triggers, taper templates, medical red-flag detection, workload trend alerts, and recommendation confidence scoring. That recommendation follows directly from the evidence landscape: the literature gives enough structure on programming variables to support stable rules, while monitoring literature repeatedly shows that noisy readiness signals are best interpreted through multi-signal models rather than ad hoc guesswork. ³³

A practical deterministic module layout would look like this:

Module	Inputs	Output
Exercise classifier	Exercise ID, variant metadata	Pattern, muscles, fatigue cost, complexity, contraindication tags
Volume engine	Session logs	Weekly direct/fractional sets by muscle and pattern
Capability engine	Performance history, RIR/RPE, optional velocity	Estimated current capability and confidence band
Readiness engine	Sleep, soreness, stress, illness, travel, cardiorespiratory strain, recent load	Local and global readiness score
Progression engine	Planned vs performed work, readiness, pain, trend model	Increase / hold / reduce / swap / add reps / add sets
Risk engine	Symptoms, pain, recent spikes, red flags	Green / yellow / red action state
Trend engine	Rolling performance and adherence	Plateau, breakthrough, inconsistency, possible under-recovery
Phase engine	Goal, training age, calendar	Template, mesocycle goal, deload / taper timing

Module	Inputs	Output
Confidence engine	Data completeness, signal agreement, novelty	Recommend / recommend with caveat / ask question

The LLM should do the opposite kinds of work. It should **explain decisions**, answer questions, summarize workouts, compare trends in plain language, personalize education, generate non-repetitive encouragement, propose clarifying questions, and translate hard coaching logic into human-friendly guidance. It should **not** decide load jumps, calculate volume, infer medical safety from vague symptoms without rules support, or choose whether a deload happens. In short: the LLM should be the voice and interpreter, not the arbiter of core coaching logic. ³⁴

A robust LLM prompt architecture should include four layers. The **base system layer** defines role, boundaries, and tone. The **session context layer** injects current workout state, recent readiness, constraints, and risk flags. The **decision layer** provides deterministic outputs in structured form, such as `progression_state`, `readiness_state`, and `recommended_action`. The **communication layer** tells the LLM how to explain the decision: reference user history, avoid generic praise, explain why, admit uncertainty when confidence is low, and ask the smallest high-value clarifying question if needed. This architecture prevents the common failure mode where the LLM hallucinates causes because it was never told what the rules engine already knew.

Atlas also needs a **confidence system**. Confidence should be computed, not hand-waved. One useful framework is to score confidence from four ingredients: data completeness, signal agreement, novelty, and risk. When the user logs load/reps/RIR consistently and the readiness signals agree, Atlas can act with high confidence. When data are sparse, signals conflict, or symptoms imply elevated risk, Atlas should shift from declarative coaching to clarification. A simple operational version would be:

Confidence state	Typical conditions	Atlas behavior
High	Complete training data, stable history, low-risk context, signal agreement	Make a direct recommendation
Medium	Minor missingness or mild signal conflict	Recommend, but explain uncertainty briefly
Low	Major data gaps, contradictory signals, novel context, symptoms, or high-risk flags	Ask targeted questions before prescribing; if symptoms are red-flagged, escalate rather than probe

This design reflects the underlying evidence reality: recovery metrics, wellness scores, and performance signals are helpful but imperfect, and single markers such as HRV are nonspecific. ³⁵

For knowledge sources, Atlas should learn from a ranked evidence stack rather than treating all “fitness content” equally.

Trust tier	Source type	Best use	Where it is strongest	Where caution is needed
Highest	Consensus statements and position stands from ACSM, NSCA, IOC, CDC-linked return-to-play guidance, Dietitians of Canada/AND/ACSM, ISSN for sports nutrition basics	Base rules and safety rails	Broad recommendations, minimum standards, population-level guidance	Often conservative, sometimes less helpful for niche advanced cases
Very high	Systematic reviews, umbrella reviews, and meta-analyses	Default evidence for programming logic	Dose-response, average effects, direction of evidence	Heterogeneity, short study durations, weak external validity to elites
High	Major review papers and textbooks	Ontology building and conceptual modeling	Definitions, frameworks, biomechanics overviews	Can lag behind newest evidence
Moderate	Leading researchers such as Stuart Phillips, Brad Schoenfeld, Jozo Grgic, Eric Helms, and colleagues whose work repeatedly appears in resistance-training reviews	Tracking frontier questions and practical interpretation	Hypertrophy, volume, protein, monitoring, autoregulation	Individual researchers can have domain blind spots; Atlas should follow findings, not personalities
Moderate	Evidence-oriented coaches and practitioners	Translating theory into implementable heuristics	Programming edge cases, athlete communication, sport-specific nuance	High risk of overgeneralization, anecdotal drift, and commercial bias
Lower	University resources and high-quality professional education sites	Explanatory content and onboarding education	Clear pedagogy and definitions	Not a substitute for primary evidence

ACSM's official position-stand process is especially valuable because it is an evidence-based organizational pronouncement rather than one lab's interpretation. NSCA position statements are similarly useful for youth and older adults. Systematic reviews then refine the rules, especially around variables like set volume, frequency, failure, rest intervals, and periodization. ³⁶

The implementation roadmap should be prioritized by how much it improves coaching quality versus engineering complexity.

Priority phase	What to build first	Why it comes first
Foundation	Exercise ontology, user profile model, workout log schema, volume engine, capability estimates, simple progression rules, safety triage	Without this, Atlas is just a chat wrapper around a logbook
Coaching core	Readiness model, fatigue state, substitution graph, session design logic, explanation layer, adherence-aware reminders	This is where Atlas starts behaving like a real coach
Intermediate intelligence	Plateau detection, deload logic, phase templates, cohort personalization, richer per-muscle analytics, confidence system	Needed to move beyond linear beginner coaching
Advanced specialization	Powerlifting peak/taper logic, bodybuilding volume landmarks, athletic power/RFD pathways, concurrent training planner, travel and heat logic	Important for advanced users but not day-one critical
Frontier layer	Wearables integration, velocity-based optional modules, causal response modeling, reinforcement learning from user outcomes	Powerful, but only after the fundamentals are trustworthy

If the goal is to build Atlas over several years, the real strategic insight is this: **world-class coaching intelligence is not mainly about knowing more facts about fitness**. It is about representing the right entities, storing the right history, separating stable rules from adaptive reasoning, and refusing to let fluent text substitute for consistent coaching logic. The strongest evidence in resistance training already supports much of the deterministic backbone. The LLM's job is to make that backbone feel intelligent, human, and personal without ever replacing it. ³⁷

¹ ¹⁹ ³⁷ American College of Sports Medicine Position Stand. ...
https://pubmed.ncbi.nlm.nih.gov/41843416/?utm_source=chatgpt.com

² ⁵ ¹³ ²⁰ ²⁹ ACSM Publishes Updated Resistance Training Guidelines
<https://acsm.org/resistance-training-guidelines-update-2026/>

³ ³¹ Monitoring the athlete training response: subjective self ... - PMC
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⁴ ¹² ³⁴ Exercise, physical activity, and self-determination theory
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